

P. VINOTH KUMAR

Post Silicon Validation/Triage/Debug Engineer

Harman Connected Services Corp India Pvt Ltd, Bangalore-India

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Summary:

- Having 7+ years of experience in Post Silicon Validation, Functional testing, Platform Validation, Sanity Testing, Stress testing, Regression Testing and Bug cycle tracking.
- Experienced in preparing Test cases, Test Execution, Test Plan, Test Strategy and Test Estimation.
- Expert in Post Silicon Validation and Debugging.
- Handled end to end validation/debug in Type -C subsystem, Storage, Power management, Sensor, Touch Module, and Debug & Trace domain.
- Specialized in Thunderbolt, Type-C, Storage & USB IO Domain.
- Hands on experience in BIOS/UEFI testing.
- Good experience in Firmware Integration and Validation (BIOS, EC)
- Comprehensive experience on All device driver's validation.
- Strong knowledge about Intel Power management, BIOS and platform booting process.
- Experience in debug Tracing NPK logs using all debug interface (E2E) for all platforms.
- Hands on Experience on debug with PythonSV using ITP and involving in collecting the Status Scope Logs, Soft & Hard Hang logs, MCA Dump, Memory dump, Reading/ Writing the values of Registers, Dumping BIOS knobs, Reading the Post Codes using the respective Python Commands.
- Extensive exposure with SDLC and STLC.
- Expert in preparing technical BKM's for future reference.
- Team leading and management skills.
- Hand on Experience in CHIP level & CARD level Debugging of Motherboards, Tablet, Smartphone's, and Laptops in L1 to L3 level.
- Knowledge in SMT and SMD schematic diagram and testing.
- Excellent communication and interpersonal skills.
- Smart worker interested to learn new technologies and achieve results of the targets within the time scale.

Technical Skill Set:

- **OS:** Windows, Linux Basic
- **SoCs:** Intel Core
- **Test Management tools:**
Phoenix, JAMA, HSDDES, Bug Assist Tool
- **Test Types:**
Functional, Regression, Firmware Test, Stress & Stability, Driver Testing
- **Script Knowledge:**
C Programming Language (Beginner)
- **Protocols:**
USB, USB Type-C, Thunderbolt, Storage
- **Debug & Hardware Tools:**
Dediprog SF600, Y2Prog, Tera Term, Putty, Beagle Analyzer, Cypress Protocol Analyzer, LeCroy protocol analyzer, Travel Logic/MSO, BSSB-DCI, USB2/3 DBC Cable, Lauterbach, Intel XDP, Type-C Power Meter, Thermal Heat Gun, JTAG based Debugging, and SW Debug tools (Windbg, etc.)
- **Hardware Knowledge:**
Good in analyzing Electricals's schematic Diagram, Testing the Signal levels in PCB, Debugging of Laptop Motherboards, Tablets, and Smartphone's, multi meter, OS Installation – Laptop, Tablet & Smartphone.

Work Experience & Responsibilities:

Senior Product Engineer

Duration: Aug 2023 – Sep 2024

Harman Connected Services Corp India Pvt Ltd, Bangalore, KA

Client Name: Intel Technology India Pvt Ltd, Bangalore, Karnataka.

Description: Developing the test plans and test content creation for new implementations enabled (EC firmware), and Device Driver Testing.

Roles and Responsibility:

- Experienced in test case content creation for EC firmware new features.
- Experienced in Requirement Analysis, Creating Test scenarios, Test cases, and Test case execution.
- Experience in Embedded Controller (EC) firmware feature testing.
- Involved in Test Case Development by converting functional requirements into test.
- Interact with domain team on frequent basis to understand the architecture details and changes (if any) to keep the team updated.
- Responsible for building knowledge and enabling team members & new on Boarding.
- Experienced in working with technical teams or equally comfortable working alone without supervision.
- Running test scenarios of Total Storage Encryption (TSE) Driver Post-Silicon Environment.
- Working on Securing data using Encryption and Decryption process for the Storage devices.
- Mainly we are working for data security. i.e., Data encryption and decryption.
- Bringing up the platform based on Configuration update received from Client.
- Attending domain forums for effective addressing of bugs & Verification.

Senior Test Engineer

Duration: Nov 2021 – Aug 2023

Wipro Limited, Bangalore, KA

Client Name: Intel Technology India Pvt Ltd, Bangalore, Karnataka.

Description: Triaging/Debugging of Type C Sub System, Storage, Power Management and Debug and Trace, (E2E) Compliance by providing necessary logs (CSME, PD, EC, PMC, ETW, NPK, BIOS, SPI, VISA Logs etc.)

Roles and Responsibility:

- Managing a team of 12 members providing end-to-end validation and debug support for USB-C sub system & Storage team.
- Handling Complete Responsible for end-to-end Bug Tracking until closure as per ETA.
- Handled single handedly, several device detections issues on hot plug / unplug and device lost issues after resume from sleep or hibernate.
- Work closely with domain team by providing necessary logs, debug information and provide probable root cause analysis for the issues.
- Logging/Tracing of different failures using Windows debug tools, Eclipse, Trace32, Beagle I2C Analyzer, Ellisys Analyzer, Aardvark analyzer, Cypress CY4500 PD protocol analyzer, Lecroy protocol analyzer, Lauterbach x86/x64 Debugger etc.
- Event viewer logs, Application logs analysis for all basic functionalities & issues.
- Assist development teams in debugging defects by providing NPK, PD, EC, PMC, ETL, ETW, BIOS, VISA Logs.
- BSOD, Soft Hang with Error Post Code issue can be Analyzes using Windbg & Hard Hang, Cat error using ITP tools.
- Experience in collecting memory Dump file and finding the failure using Windbg Debugging tool.
- Collection and analysis of debug logs using PythonSV.
- Verification of the engineering fixes and initiating for official fix to be integrated in next official release.
- Unblocking the execution blockages and help to run the execution smoothly.
- Responsible for providing a technical guidance to the team for feature enablement.
- Handling the client meetings whenever necessary.

Senior Validation Engineer

Duration: Apr 2019 - Oct 2021

Validation Engineer

Duration: May 2016 - March 2018

UST Global, Bangalore, KA

Client Name: Intel Technology India Pvt Ltd, Bangalore, Karnataka.

Description: Validating/Triaging of Type C Sub System, Debug and Trace, Storage, Power Management, Thermal Management, Touch Module, and Sensor module.

Roles and Responsibility:

- Setup the board as per the hardware configuration matrix for the release consisting of platforms, peripherals, add-on cards, firmware, BIOS etc.
- Involving in platform validation, triage, isolation, Enablement & debugging with debug tools.
- Execution of various validating methods like Driver Validation (MiniBAT), Regression, Functional, and Stress testing with Intel platforms.
- Validation across Type C Sub System, Debug and Trace, Storage, Power Management, Thermal Management, Touch Module, and Sensor module.
- Validating various Type-C / Thunderbolt Dock behind storage and display related etc.
- Strong Knowledge in RED PVT/ISD tool/ PythonSV Nightly installation and tracing the logs.
- Good experience in Third Party Ingredients Integration and Validation. (Touch Panel, (SPI/I2C), Touch Pad (I2C), BayHUB, PCIe, Fingerprint Sensor, NVMe Firmware flashing etc.)
- Knowledge/Hands-on Storage domain such as various RAID levels testing using different AHCI, SATA, SSD, NVMe M.2 Drives.
- Good Knowledge in RAID, RST & VMD concepts.
- Expertise in performance and stress testing on SATA NVMe, Intel SSD, M.2 PCIe NAND drives and thermal analysis on all core series.
- Knowledge of NVMe/SSD/HDD and UFS, eMMC interfaces and working speed and specifications.
- Good knowledge on discrete Graphic Card, system benchmarking and other graphics tools.
- Collecting stress and stability information for SoC, GPU, and CPU systems.
- Validating performance test on high-speed IOs, including PCIe, DGx card, Nvidia graphics card & AMD graphics card, system benchmarks with different API's and other graphics tools.
- Collecting Lauterbach JTAG logs using Trace32 and openIPC logs using PVT.
- Collecting windbg logs with UART, Mini or Micro USB, LAN cable, Type A to Type A debug cable.
- Tools Proficiency- Intel System Scope, Intel Cycling, CrashLog, Putty, MSR Read and Write, IO meter, Crystal Disk Mark, DigiInfo, Thermal analysis tools etc.

Debugging Engineer

Duration: Sep 2018 - April 2019

Flextronics Technologies India Private Limited, Chennai, TN

Diagnostic Engineer

Duration: March 2018 - Sep 2018)

Rising Stars Mobiles India Private Limited (Foxconn Technology), Chennai, TN

Description: Debugging and analyzing various SMT failures of mobile PCBA Devices like Xiaomi, Motorola, Yuwei, and Jio, up to L3 and L4 level & also Managed 40 members team Shift activities.

Roles and Responsibility:

- Debugging of mobile PCBA Devices like Xiaomi, Motorola, Yuwei, and Jio using CMW500 Tester.
- Debugging of all PCBA Functional Testing Failures of Calibration, Radio Frequency, and Download test Fail using CMW500.
- Performed mobile PCBA-BGA rework and rebelling. (Fine Tech.Inde & BT3100).
- Analyzing the failed PCBA and assembled mobiles which are received from production line and providing quick feed backs to stop the failure and reduce MFR.
- Responsible for providing a technical guidance to the team for troubleshooting.
- Responsible for preparing FAR Reports (Failure Analysis Report) for the Yield losses in the production lines of Mobile Products.
- Installation and Configuration of All Online/offline Mobile Software.

- Exposure to multiple repair platforms requiring advanced soldering and the use of soldering equipment.
- Maintains All the Report of Service on Daily, Weekly and Monthly Basis.
- WIP clearness & awarding within a TAT.

Debug Testing Engineer

Duration: Nov 2011 - Feb 2015)

Regeneresis India Pvt Ltd, Chennai, TN

Description: Testing/Debugging of various Laptop motherboard like Lenovo, Dell, ASUS, HP, Acer etc.

Roles and Responsibility:

- Debugging of Laptop Motherboards, Tablets, Smart phone, and Desktop.
- Testing all types of Dell, HP, Acer, Lenovo & Desktop mother boards.
- Assembling & disassembling all types of Notebook & Desktops.
- Diagnosing of Tablets, upgrading firmware, Solving the issues in chip level (L3 issues).
- Visual inspection of PCB BOARD. (Like track damage, rooting & physical damages).
- Functional testing, Quality checking, SMD rework.
- Card level and Chip level (L3) Debugging of Complete Laptop.
- OS Installation & Troubleshooting.

Educational Qualifications:

Bachelor of Electronics and Communication Engineering (ECE),
Anna University in the year 2011,
Percentage: 76%

Diploma in Electronics and Communication Engineering (ECE),
DOTE in the year 2008,
Percentage: 87%

Declaration:

I hereby declare that the information given above is true and correct to best of my knowledge.

Place: Bangalore

Vinoth Kumar